AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: GEN26-AOH-I011

Team Name: HACK ORBIT

Domain: Digital Technologies, AI & Cyber Security

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Presentation Outline

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Problem Statement

Basic Network Packet Sniffer & Analyzer[cite: 1]
Everything we do online is chopped into tiny packets. Today it is hard for beginners to see them. We want a free tool that catches those packets and shows who is talking to whom, in plain English – like a CCTV camera for your Wi-Fi.[cite: 1]

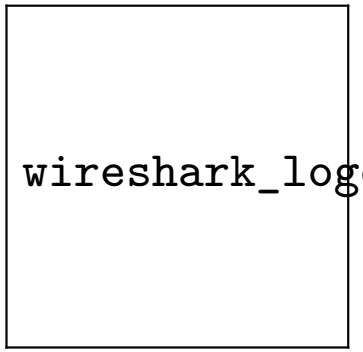
packet_inspection.png

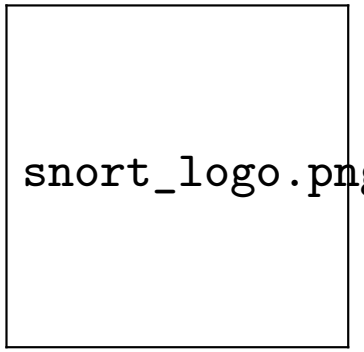
Problem Understanding

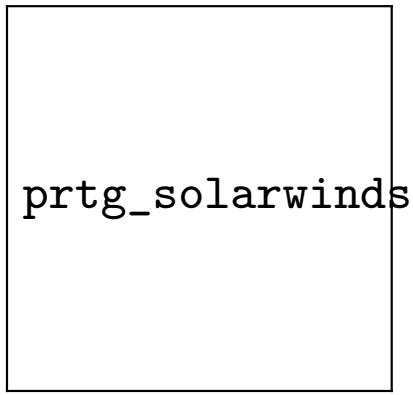
- **Who experiences this problem?** Network admins, security analysts, students and developers – and also ISPs, teachers and anyone whose video call has ever frozen.[cite: 1]
- **What causes the problem?** Packets are a wall of numbers. Most tools are made for experts and show raw data, not simple answers, and capture, analysis and charts live in separate tools.[cite: 1]
- **How does it affect them?** Fixes take longer, attacks slip by unnoticed, bandwidth gets wasted and learners never see how networks really work.[cite: 1]

Existing Gaps & Challenges

- **Powerful tools are too complex:** Wireshark throws hundreds of fields at a beginner, tcpdump is just text with no charts, and Snort / Suricata take real expertise to set up.[cite: 1]
- **Simple or paid options fall short:** ping / traceroute say something is wrong, not why. SolarWinds and PRTG are great but pricey and built for big firms.[cite: 1]
- **Main challenge that remains unsolved:** nobody offers a free, beginner-friendly tool that captures, explains, charts and warns, all in one place.[cite: 1]



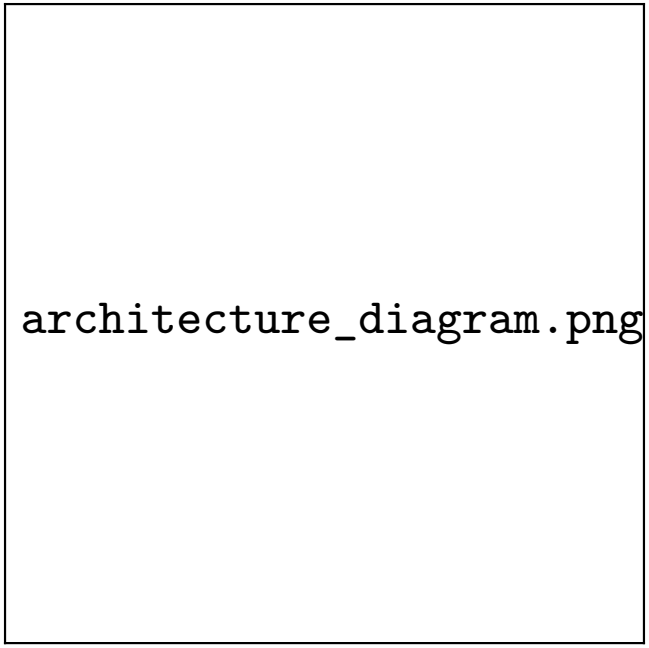




Our Proposed Solution

- **Core idea:** a network packet sniffer and analyzer that catches live traffic and shows it on a clean dashboard – who, where, which protocol, how big.[cite: 1]
- **How it works:** live capture, then filters for TCP, UDP, ICMP and DNS, then summaries and charts, plus friendly alerts when it spots port scans or traffic spikes.[cite: 1]
- **Main value to users:** beginners can actually read their traffic, slow or broken networks get fixed faster, and results export to CSV and PCAP.[cite: 1]

Technical Approach



Packets are captured live, analysed and filtered, then shown on one dashboard with alerts and exports.[cite: 1]

Tech Stack[cite: 1]

Frontend: Dashboard with charts (Matplotlib)

Backend: Python + Scapy packet capture

Database: Pandas for analysis; CSV / PCAP export

Hardware/Libraries: Laptop (8 GB RAM), Npcap / libpcap

Innovation & Uniqueness

- **What is new or different?** one tool that captures, explains, charts and warns, instead of juggling several separate tools.[cite: 1]
- **Why is this approach suited to the problem?** it shows traffic as simple answers, not raw data, so beginners, students and small teams can use it on a regular laptop.[cite: 1]
- **Key differentiator:** free and beginner-friendly, with built-in alerts for port scans and spikes – easy to use after one demo.[cite: 1]

Feasibility & Scalability

- **Feasibility:** a regular laptop with 8 GB RAM, free tools (Python, Scapy, Npcap / libpcap, Pandas, Matplotlib), a team of four and a faculty mentor. Wireshark is used to double-check our results.[cite: 1]
- **Scalability:** start on a small test network and a second device, then grow by adding more protocol filters and alert rules.[cite: 1]
- **Risks:** we capture only on networks we are allowed to monitor, and use sample packet files for demos.[cite: 1]

Cost for Product (Optional)

- **Development or hardware cost:** free software (Python, Scapy, Npcap / libpcap, Pandas, Matplotlib) on a laptop the team already has.[cite: 1]
- **Hosting or maintenance cost:** none expected – the tool runs on a local laptop.[cite: 1]
- **Estimated total:** minimal – no expensive software needed.[cite: 1]

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1:** Network admin. Wi-Fi acts up – the admin sees which device and app is eating the bandwidth and fixes it quickly.[cite: 1]
- **Use case 2:** Security analyst. Port scans and traffic spikes are flagged by friendly alerts, so suspicious activity is caught early.[cite: 1]
- **Use case 3:** Student or teacher. Watch a network work live in the lab, with protocol charts and sample packet files, instead of just reading about it.[cite: 1]

Impacts and Benefits

- **Benefit to the target users:** anyone can read their network traffic; problems get fixed faster and threats get caught earlier.[cite: 1]
- **Social and economic impact:** no expensive software needed, and learners pick up real networking and security skills.[cite: 1]
- **Expected measurable outcome:** filters and charts work, alerts fire, data exports to CSV and PCAP, and it beats doing it by hand.[cite: 1]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.4	Quality Education	Students learn networ security by seeing live building practical tech skills.[cite: 1]
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Helps keep networks r secure by catching slo threats early.[cite: 1]

Individual Member Contribution

Member	Contribution or responsibility
Kavish Rahav D P	Team Lead & Backend Developer. Builds the engine that catches packets and ties everything together.[cite: 1]
Karthikeyan K	Network & Security Analyst. Works out what the traffic means and sets the alert rules.[cite: 1]
Ratchitha Kathiravan Vedhavalli	Frontend / UI Developer. Designs the dashboard you actually see and use.[cite: 1]
Jeyanthan S	Tester & Documentation Lead. Tests everything, checks it against Wireshark, writes it all up.[cite: 1

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